

# Noise Scale as C & G Harmonic Displacement

## Richardson Extrapolation Centred on the Perfect Harmonic Reference

*Theoretical development for Universal Harmonic Theory*

**Scientific status:** The Richardson mathematics is established. The interpretation of noise as displacement from a perfect C & G harmonic state is a proposed, testable extension and is not an established law of physics.

### 1. Core Definition

Within this framework, the noise scale is not treated only as an arbitrary amplification factor. It is redefined as a measurable displacement from a perfect C & G harmonic reference condition. The reference ratio is the perfect fifth:

$$\omega_G / \omega_C = 3 / 2$$

At perfect harmonic balance, the normalised harmonic scale is defined as unity:

$$\lambda_{CG} = 1$$

The normalised C & G harmonic scale is therefore

$$\lambda_{CG} = (\omega_G / \omega_C) / (3/2) = 2\omega_G / 3\omega_C$$

Values above or below unity represent displacement away from the proposed equilibrium ratio.

### 2. Harmonic Displacement Coordinate

The direct displacement from the perfect harmonic reference is

$$\delta_{CG} = \lambda_{CG} - 1$$

A logarithmic coordinate gives a symmetric measure of upward and downward departures:

$$\chi_{CG} = \ln(\lambda_{CG}) = \ln(2\omega_G / 3\omega_C)$$

At exact balance,

$$\chi_{CG} = 0$$

The logarithmic form is useful because reciprocal deviations from the reference ratio have equal magnitude and opposite sign.

### 3. Harmonic Noise and Coherence

Define the magnitude of harmonic noise as the absolute displacement in logarithmic ratio space:

$$N_{CG} = |\chi_{CG}|$$

Therefore,

$$N_{CG} = 0 \text{ at perfect C \& G balance}$$

A bounded harmonic-coherence function may be introduced as

$$H_{CG} = \exp[-N_{CG}^2 / (2s_{CG}^2)]$$

where  $s_{CG} > 0$  is a dimensionless resonance-width parameter. This gives

$$H_{CG} = 1 \text{ when } N_{CG} = 0$$

$$0 < H_{CG} < 1 \text{ when } N_{CG} > 0$$

## 4. Richardson Extrapolation About Harmonic Equilibrium

Conventional Richardson extrapolation is centred on a formal zero-noise coordinate. Under the C & G formulation, the physical reference is  $\lambda_{CG} = 1$ . The observable is expanded about that point:

$$E(\lambda_{CG}) = E(1) + a_1(\lambda_{CG} - 1) + a_2(\lambda_{CG} - 1)^2 + \dots$$

Using  $\delta_{CG} = \lambda_{CG} - 1$ , this becomes

$$E(\delta_{CG}) = E(0) + a_1\delta_{CG} + a_2\delta_{CG}^2 + \dots$$

Here  $E(0)$  means zero harmonic displacement, not a zero frequency ratio:

$$E(0) \equiv E(\delta_{CG} = 0) = E(\lambda_{CG} = 1)$$

The central equivalence is

$$\text{zero harmonic noise} \Leftrightarrow \delta_{CG} = 0 \Leftrightarrow \lambda_{CG} = 1 \Leftrightarrow H_{CG} = 1$$

## 5. General Harmonic Richardson Conditions

Suppose measurements are obtained at  $n$  harmonic displacements  $\delta_1, \delta_2, \dots, \delta_n$ . The perfect-balance value is estimated by

$$\hat{E}_{CG}^* = \sum_{j=1}^n \gamma_j E(\delta_j)$$

The coefficients satisfy

$$\sum_{j=1}^n \gamma_j = 1$$

$$\sum_{j=1}^n \gamma_j \delta_j^k = 0, \quad k = 1, 2, \dots, n-1$$

Equivalently, since  $\delta_j = \lambda_{CG,j} - 1$ ,

$$\sum_{j=1}^n \gamma_j (\lambda_{CG,j} - 1)^k = 0$$

These conditions preserve the equilibrium term while cancelling the selected orders of harmonic displacement.

## 6. First-Order Example

Take two measurements at

$$\lambda_{CG,1} = 1 + \Delta$$

$$\lambda_{CG,2} = 1 + 3\Delta$$

Their displacement coordinates are

$$\delta_1 = \Delta, \quad \delta_2 = 3\Delta$$

For the linear model

$$E(\delta) = E_{\text{CG}}^* + a_1 \delta$$

the measured values are

$$E(\Delta) = E_{\text{CG}}^* + a_1 \Delta$$

$$E(3\Delta) = E_{\text{CG}}^* + 3a_1 \Delta$$

Solving for the perfect-balance value gives

$$E_{\text{CG}}^* = [3E(\Delta) - E(3\Delta)] / 2$$

or, in terms of the normalised harmonic scale,

$$E_{\text{CG}}^* = [3E(1 + \Delta) - E(1 + 3\Delta)] / 2$$

## 7. Quadratic Harmonic Extrapolation

For three displacement levels

$$\delta = [\Delta, 3\Delta, 5\Delta]$$

the second-order Richardson estimator is

$$E_{\text{CG}}^* = [15E(\Delta) - 10E(3\Delta) + 3E(5\Delta)] / 8$$

In terms of the harmonic scale,

$$E_{\text{CG}}^* = [15E(1 + \Delta) - 10E(1 + 3\Delta) + 3E(1 + 5\Delta)] / 8$$

This cancels first- and second-order departures from the perfect C & G reference when the local polynomial model is valid.

## 8. Inclusion of the Fourth Dimension

Because harmonic balance can evolve through spacetime, the harmonic scale is written as a field:

$$\lambda_{\text{CG}} = \lambda_{\text{CG}}(x^u), \quad x^u = (ct, x, y, z)$$

The displacement field is

$$\delta_{\text{CG}}(x^u) = \lambda_{\text{CG}}(x^u) - 1$$

The observable becomes

$$E = E[\delta_{\text{CG}}(x^u), x^u]$$

The spacetime-dependent harmonic extrapolation is

$$\hat{E}_{\text{CG}}^*(x^u) = \sum_{j=1}^n \gamma_j E[\delta_{\text{CG},j}(x^u), x^u]$$

For the quadratic sequence,

$$\hat{E}_{\text{CG}}^*(x^u) = [15E(\Delta, x^u) - 10E(3\Delta, x^u) + 3E(5\Delta, x^u)] / 8$$

The equilibrium condition is therefore

$$\delta_{CG}(x^u) \rightarrow 0$$

or equivalently

$$\lambda_{CG}(x^u) \rightarrow 1$$

## 9. Time-Dependent Harmonic Evolution

Writing the fourth coordinate explicitly, the scale may be expressed as  $\lambda_{CG}(t, x, y, z)$ . Along a fixed worldline, the time-dependent equilibrium estimate is

$$\hat{E}_{CG}^*(t) = \sum_{j=1}^n \gamma_j E[\delta_{CG,j}(t), t]$$

The local expansion contains both harmonic-displacement and temporal terms:

$$E(\delta, t) = E(0, t_0) + E_{\delta} \delta + E_t \Delta t + \frac{1}{2} E_{\delta\delta} \delta^2 + E_{\delta t} \delta \Delta t + \frac{1}{2} E_{tt} (\Delta t)^2 + \dots$$

Ordinary one-variable Richardson cancellation removes selected powers of  $\delta$ , but it does not automatically remove temporal drift  $E_t \Delta t$  or the mixed term  $E_{\delta t} \delta \Delta t$ . Measurements should therefore be time aligned or treated with a multidimensional extrapolation.

## 10. Multidimensional Harmonic Extrapolation

Let  $\tau = (t - t_0)/T_0$  be a dimensionless time displacement. A joint expansion is

$$E(\delta, \tau) = \sum_p \sum_q a_{pq} \delta^p \tau^q + R$$

A multidimensional estimator is

$$\hat{E}(0, 0) = \sum_{j=1}^N w_j E(\delta_j, \tau_j)$$

with constraints

$$\sum_j w_j = 1$$

$$\sum_j w_j \delta_j^p \tau_j^q = 0 \text{ for each cancelled pair } (p, q) \neq (0, 0)$$

For a first-order model that also cancels the mixed term, one imposes

$$\sum_j w_j \delta_j = 0$$

$$\sum_j w_j \tau_j = 0$$

$$\sum_j w_j \delta_j \tau_j = 0$$

## 11. Unified Interpretation

The proposed conceptual identity is

$$\text{noise} = \text{departure from harmonic equilibrium}$$

In operational form,

$$N_{CG}(x^u) = | \ln[2\omega_G(x^u) / 3\omega_C(x^u)] |$$

The perfect C & G state is

$$N_{CG} = 0, \quad \lambda_{CG} = 1, \quad \delta_{CG} = 0, \quad H_{CG} = 1$$

The extrapolation limit is

$$E_{CG}^*(x^u) = \lim_{\{\lambda_{CG} \rightarrow 1\}} E(\lambda_{CG}, x^u)$$

Thus the conventional zero-noise coordinate and the harmonic equilibrium coordinate are compatible when the displacement variable is used:

$$\lambda_{\text{ordinary}} \rightarrow 0$$

$$\lambda_{CG} \rightarrow 1$$

$$\delta_{CG} = \lambda_{CG} - 1 \rightarrow 0$$

## 12. Statistical Consequence

The harmonic reinterpretation does not remove the Richardson variance penalty. If the measurements have equal independent variance  $\sigma^2$ , then

$$\text{Var}(\hat{E}_{CG}^*) = \sigma^2 \sum_j \gamma_j^2$$

For the quadratic coefficients 15/8, -10/8 and 3/8,

$$\sum_j \gamma_j^2 = 334/64 = 5.21875$$

Hence

$$\text{Var}(\hat{E}_{CG}^*) = 5.21875 \sigma^2$$

and the corresponding standard-deviation amplification is

$$\sigma_{\text{ext}} / \sigma = \sqrt{5.21875} \approx 2.284$$

## 13. Testable Status

The equation structure provides a mathematically consistent way to centre Richardson extrapolation on a defined harmonic reference. Scientific validation requires experimental definitions for  $\omega_C$  and  $\omega_G$ , a reproducible method for controlling  $\lambda_{CG}$ , a specified observable  $E$ , and evidence that the harmonic formulation predicts residual error or physical behaviour more accurately than conventional noise models.

The proposed hypothesis may be stated as

$H_0$ : harmonic displacement adds no predictive information

$H_1$ : residual error varies reproducibly with  $\delta_{CG}$  or  $H_{CG}$

Until such evidence exists, the C & G interpretation should be identified as a theoretical extension rather than an experimentally established law.

### CENTRAL HARMONIC LIMIT

$$\lambda_{CG} = 1 \Leftrightarrow \delta_{CG} = 0 \Leftrightarrow N_{CG} = 0 \Leftrightarrow H_{CG} = 1$$